

CBSE Test Paper 05
Chapter 5 Laws of Motion

1. Two bodies of masses 10 kg and 20 kg respectively kept on a smooth, horizontal surface are tied to the ends of a light string. a horizontal force $F = 600 \text{ N}$ is applied to the 10 kg mass so as to pull it. What is the tension in the string? **1**
 - a. 450 N
 - b. 300 N
 - c. 400 N
 - d. 370 N

 2. A train runs along an unbanked circular track of radius 30 m at a speed of 54 km/h. The mass of the train is 106 kg. What is the angle of banking required to prevent wearing out of the rail? ($g = 10 \text{ m/s}^2$) **1**
 - a. 37°
 - b. 32°
 - c. 27°
 - d. 42°

 3. A truck starts from rest and accelerates uniformly at 2.0 m s^{-2} . At $t = 10 \text{ s}$, a stone is dropped by a person standing on the top of the truck (6 m high from the ground). What is the magnitude of velocity (in m s^{-1}) of the stone at $t = 11 \text{ s}$? (Neglect air resistance.) **1**
 - a. 22.4 m/s
 - b. 19.6 m/s
 - c. 18.5 m/s
 - d. 21.0 m/s

 4. A man of mass 70 kg stands on a weighing scale in a lift which is moving upwards with a uniform speed of 10 m s^{-1} , what would be the reading on the scale? **1**
 - a. 75 kg
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- b. 70 kg
c. 35 kg
d. 105 kg
5. A rocket with a lift-off mass 20,000 kg is blasted upwards with an initial acceleration of 5.0 m s^{-2} . Calculate the initial thrust (force) of the blast. **1**
- a. 300000 N
b. 350000 N
c. 378000 N
d. 365000 N
6. Explain why passengers are thrown forward from their seats when a speeding bus stops suddenly. **1**
7. A car and a truck have the same momentum. Which of the two have greater speed and why? **1**
8. Name the factor on which coefficient of friction depends. **1**
9. A constant retarding force of 50 N is applied to a body of mass 30 kg moving initially with a speed of 18 ms^{-1} . How long does the body take to come to a halt? **2**
10. A nucleus is at rest in the laboratory frame of reference. Show that if it disintegrates into two smaller nuclei the products must move in opposite directions. **2**
11. What is kinetic friction? Is it self-adjusting? **2**
12. A hammer of mass 1 kg strikes on the head of a nail with a velocity of 10 ms^{-1} . It drives the nail 1 cm into a wooden block. Calculate the force applied by the hammer and the time of impact. **3**
13. A pebble of mass 0.05 kg is thrown vertically upwards. Give the direction and magnitude of the net force on the pebble, **3**
- a. during its upward motion,
b. during its downward motion,
c. at the highest point where it is momentarily at rest. Do your answers change if the
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pebble was thrown at an angle of 45° with the horizontal direction? Ignore air resistance.

14. a. State impulse-momentum theorem.
b. A ball of mass 0.1 kg is thrown against a wall. It strikes the wall normally with a velocity of 30 m/s and rebounds with a velocity of 20 m/s . Calculate the impulse of the force exerted by the ball on the wall. **3**
15. Two bodies A and B of masses 5 kg and 10 kg in contact with each other rest on a table against a rigid wall. The coefficient of static friction, μ_s between the bodies and the table is 0.15 . A force of 200 N is applied horizontally to A. What are
- the reaction of the partition
 - the action-reaction forces between A and B?

What happens when the wall is removed? Does the answer to (b) change, when the bodies are in motion? Ignore the difference between μ_s and μ_k . (Given, $g = 10\text{ m/s}^2$) **5**

